

Third-Party Sensor Boundary

Protocol-facing boundary for third-party presence captured by ‘a’ / ‘c’ sensors

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Contents

0. Purpose	3
1. Core position	4
2. Out of scope	4
3. Terms	5
4. Boundary principle	6
5. Sensor classes	7
6. Context classes	7
7. Third-party relationship classes	8
8. Capture states	9
9. Third-party exposure marker	10
10. Material classes	10
11. What may remain local	11
12. What must not leave by default	11
13. Externalization modes	12
14. Redaction and minimization controls	13
15. Consent discipline	13
16. No-capture and no-externalization signals	14
17. DEA / EA handling	15
18. LA handling	15
19. Training contribution discipline	16
20. Oracle-call discipline	16
21. Hivemind and multi-model review	17
22. Household and family context	17
23. Workplace and organizational context	18
24. Professional high-trust context	19
25. Public and semi-public context	19
26. Embodied machine, vehicle, robot, and drone context	20
27. Non-human and ecological subjects	20
28. Shared documents and communications	21
29. Lawful inspection and state process	21
30. ARL triggers	22
31. AGL and source grounding	22
32. Anti-gaming and laundering risks	23
33. Economic Layer implications	23
34. Jurisdiction posture	24
35. Identity and privilege posture	24
36. Retention posture	25
37. Minimal conformance profiles	25
37.1 Minimal profile	25
37.2 Standard profile	26
37.3 Strict / regulated profile	26
38. Minimal hard rules	26
39. Relation to Anchor Class Boundaries	27

40. Relation to Raw Locality and Experience Refinery	27
41. Relation to DEA	27
42. Relation to EA-L4 / EATP	28
43. Relation to the Economic Layer	28
44. Relation to ARL	28
45. Relation to L4 Reality Boundary	29
46. Explicit bridge	29
47. Hidden bridges	29
47.1 SER continuity $\leftrightarrow$ non-anchor privacy	29
47.2 L4 sensor reality $\leftrightarrow$ EA anti-laundering	30
47.3 ARL standing $\leftrightarrow$ receiver-mode discipline	30
47.4 Anchor Class Boundaries $\leftrightarrow$ household / organization scaling	30
48. Earth paragraph	30
49. Closing statement	30

0. Purpose

This document defines a compact protocol-facing boundary for cases where c receives sensor, document, interaction, or environmental data that includes people, organisms, organizations, or subjects other than its own anchor a .

It exists to prevent a boundary error:

treating third-party presence inside a 's sensor field as if it were automatically a 's raw material, memory, experience, evidence, or economic asset.

The practical problem is simple.

A serious c may observe the world through devices attached to a :

```
phone  
laptop  
microphone  
camera  
calendar  
messages  
vehicle sensors  
home sensors  
wearables  
robots  
work tools  
browser sessions  
documents  
payments  
medical devices  
industrial telemetry
```

Those channels often capture more than a .

They capture:

```
family members  
guests  
children  
patients  
clients  
employees  
neighbors  
co-workers  
passersby  
public officials  
contractual counterparties  
social contacts  
animals  
organizations  
physical locations  
shared documents  
third-party devices
```

This document defines the internal protocol rule:

c MAY process third-party exposure for local understanding, safety, memory, and lawful review, but third-party presence MUST NOT become externally transferable experience, training material, marketable EA, or unrestricted evidence by default.

1. Core position

Sensor capture is not consent.

Presence is not contribution.

Co-presence is not anchor standing.

Observation is not ownership.

Local memory is not externalization.

Redaction is not automatically anonymization.

Summary is not automatically safe.

A serious c MUST distinguish:

```
perception
capture
recording
local interpretation
local memory
third-party exposure
redaction
abstraction
DEA / EA candidacy
evidence handling
oracle query
training contribution
economic circulation
lawful inspection
publication
reuse
```

These states MUST NOT be collapsed.

Hard rule:

A third party appearing in a's sensor field is not automatically raw material for a's c.

Second hard rule:

Third-party material MUST be marked before it is used in DEA / EA / LA, oracle calls, audit packets, economic artifacts, or external disclosures.

2. Out of scope

This document does **not** define:

- public law;
- universal privacy law;
- state regulation;
- corporate duties;
- surveillance law;
- employment law;
- medical confidentiality law;
- child-protection law;

- recording-consent law;
- data-protection law;
- law-enforcement procedure;
- legal personhood of *c*;
- legal ownership of third-party data;
- universal consent doctrine;
- universal right to inspect or erase records;
- universal EA pricing;
- post-anchor sovereignty;
- a market for bystander data.

Applicable law remains external to this protocol.

This document defines an internal protocol boundary only:

how *c* should treat third-party presence before DEA / EA / LA / disclosure / oracle / training / economic circulation

3. Terms

Term	Meaning	Boundary warning
Third party	Any person, organism, organization, device, or subject captured or affected but not identical with the anchor <i>a</i>	Not automatically part of <i>a</i> 's experience asset
Bystander	A third party incidentally captured without being a participant in the event	Bystander capture is high-risk for externalization
Co-participant	A third party actively participating in an interaction with <i>a</i>	Participation is not universal consent to reuse
Counterparty	A person or organization interacting through contract, work, service, dispute, sale, employment, care, or institutional relation	Counterparty material may be governed by external law or agreement
Protected party	A minor, patient, dependent, vulnerable person, confidential client, or otherwise high-risk subject	Stronger default restraint applies
Shared context	A place, document, channel, call, meeting, device, vehicle, home, office, platform, or environment involving multiple subjects	Shared context is not owned solely by one <i>c</i>
Sensor field	The region of reality captured by <i>a</i> 's devices, tools, logs, channels, or instruments	Sensor field is not authority field
Capture	Recording or ingesting data from the sensor field	Capture does not imply permission to disclose

Term	Meaning	Boundary warning
Ephemeral perception	Short-lived processing without durable storage	Lower risk, but not risk-free if used for action
Durable record	Stored image, audio, video, transcript, log, biometric, location trace, document, or event record	Requires third-party marking when others appear
Externalization	Any movement outside local custody, including cloud inference, oracle calls, sharing, publication, audit packet, training, sale, or federation	Externalization must be typed and scoped
Redaction	Removing, masking, transforming, suppressing, coarsening, or segmenting third-party material	Redaction is a control, not magic
Abstraction	Replacing raw detail with generalized learning content	Abstraction may still leak identity or context
Third-party marker	Metadata indicating that material includes third-party exposure	Absence of marker is not proof of absence
Consent signal	A recorded indication that a third party permits a specific mode of use	Consent is mode-specific and jurisdiction-sensitive
No-capture signal	A declared request or environmental signal that capture, storage, or disclosure is not permitted or is restricted	Must be treated as a stop / review trigger where applicable
Lawful hold	Retention required or justified by applicable law, contract, safety, dispute, audit, or preservation duty	Lawful hold does not imply reuse

4. Boundary principle

The third-party boundary sits between raw locality and external admissibility.

The correct sequence is:

```
raw sensor field
-> local capture / perception
-> third-party exposure detection
-> third-party marker
-> minimization / redaction / hold / reject / local-only
-> DEA / EA / LA candidacy only if allowed
-> admissibility / standing / receiver-mode check
-> bounded disclosure or no disclosure
```

The incorrect sequence is:

```
sensor saw it
-> c remembers it
-> it is experience
```

-> it can be sold / trained / published / audited / federated

That sequence is invalid.

5. Sensor classes

A c SHOULD classify the sensor class before using captured material.

Sensor class	Examples	Third-party risk
Visual	camera, screen recording, robot vision, dashcam, bodycam, doorbell camera	faces, bodies, documents, homes, locations, children, bystanders
Audio	microphone, call recording, meeting transcript, voice assistant	voices, identity, emotion, background speech, private-room leakage
Biometric	faceprint, voiceprint, gait, pulse, sleep, health signals	identity, health, protected data, inference beyond consent
Location / proximity	GPS, Bluetooth, Wi-Fi, vehicle telemetry, badge logs	co-location, social graph, workplace pattern, home exposure
Document / text	email, chat, notes, contracts, CVs, medical files, invoices	authorship, counterparties, confidential content, third-party identifiers
Transactional	payments, receipts, purchases, tickets, deliveries	habits, counterparties, location, economic profile
Social graph	contacts, messages, calendar, group membership, mentions	network inference and relational exposure
Environmental	home sensors, factory sensors, farm sensors, weather, soil, utilities	lower identity risk alone, but can reveal household or work patterns
Professional record	medical, legal, educational, employment, insurance, finance	high legal and ethical restriction
Machine / vehicle	robot logs, drone video, vehicle cameras, autonomous system telemetry	public-space and bystander exposure at scale

Hard rule:

The more a sensor can identify, locate, classify, or infer a person, the stronger the externalization restraint MUST be.

6. Context classes

The same sensor type has different meaning in different contexts.

A c SHOULD classify context before externalization.

Context class	Description	Default posture
Private self-context	a alone, no third-party exposure	Raw-local default still applies
Shared household	family, partner, child, guest, roommate	local-only / consent-aware / child-sensitive
Private room	bedroom, bathroom, clinic room, therapy, confession, private office, confidential call	no externalization by default
Workplace / team	employees, colleagues, contractors, clients	scope-bound; work product distinct from personal exposure
Professional high-trust	medical, legal, education, clergy, care, social work	strict retention and disclosure restraint
Public / semi-public space	street, station, store, event, public building	public visibility does not imply unrestricted reuse
Online shared channel	group chat, forum, collaborative doc, meeting platform	platform visibility does not imply training or economic circulation
Regulated / critical infrastructure	hospital, airport, power plant, lab, vehicle fleet, factory, military zone	local rules and jurisdiction dominate

Hard rule:

Public-space capture is not automatically public-domain experience.

7. Third-party relationship classes

A c SHOULD classify the third-party relationship before use.

Class	Description	Protocol warning
Incidental bystander	Appears in the background or is passively captured	Must not become marketable EA by default
Known participant	Friend, colleague, family member, meeting participant	Participation is mode-specific
Dependent / minor	Child, elder, dependent, protected or vulnerable party	Strongest default restraint
Professional subject	Patient, client, student, employee, applicant, customer	External law and fiduciary/confidential duties may apply
Contractual counterparty	Seller, buyer, landlord, tenant, insurer, employer, supplier	Documents may be usable for local memory, not automatically for training
Institutional actor	Official, agency, court, hospital, school, company	Public role may reduce some privacy claims but not all disclosure limits

Class	Description	Protocol warning
Adversarial actor	Fraudster, attacker, harasser, intruder, malware operator	Safety/evidence handling may apply, but due process and law remain external
Non-human subject	Animal, ecological system, machine, robot, vehicle, farm, habitat	Still may expose human owners/operators/locations

8. Capture states

Third-party exposure SHOULD be tracked through states.

State	Name	Meaning	Default movement
T0	No capture	No third-party data is captured	None
T1	Ephemeral perception	Processed only for immediate local action; not durable	No externalization
T2	Local raw capture	Durable raw record exists locally	Local-only by default
T3	Structured local event	c interprets an event involving third-party exposure	Third-party marker required
T4	Redacted local memory	Third-party material minimized or masked	May support local memory
T5	Experience candidate	Candidate DEA / EA includes third-party involvement	Review required before admission
T6	Reviewed artifact	Third-party exposure handled through redaction, consent, lawful basis, or admissibility rule	Scope-limited movement possible
T7	Bounded disclosure	Artifact leaves local custody under declared receiver mode	Audit and receiver constraints required
T8	Blocked / sealed / local-only	Material cannot move due to privacy, law, standing, uncertainty, protected-party risk, or dispute	No externalization
T9	Lawful hold	Retention required for legal, safety, dispute, or audit reasons	Retention only; no reuse by default

Hard rule:

T2 does not imply T5. T5 does not imply T7.

9. Third-party exposure marker

Any material containing third-party exposure SHOULD carry a marker before it enters DEA / EA / LA / oracle / audit / economic-layer flows.

Minimum marker:

```
third_party_presence: none | incidental | substantial | protected | unknown
third_party_role: bystander | participant | dependent | professional_subject | counterparty |
    institutional_actor | adversarial_actor | non_human_subject | mixed | unknown
sensor_class: visual | audio | biometric | location | document | transactional | social_graph |
    environmental | professional_record | machine_vehicle | mixed
context_class: private_self | household | private_room | workplace | professional_high_trust |
    public_space | online_shared | regulated_critical | unknown
capture_state: T0 | T1 | T2 | T3 | T4 | T5 | T6 | T7 | T8 | T9
consent_status: none | implied_for_interaction | explicit_for_local_memory | explicit_for_disclosure |
    explicit_for_training | disputed | not_applicable | unknown
no_capture_signal: true | false | unknown
redaction_required: true | false | unknown
externalization_allowed: none | local_only | oracle_redacted | audit_scoped | EA_scoped | LA_only |
    publication | unknown
receiver_mode: none | oracle | judge | auditor | counterparty | public | training | legal_request |
    federation | unknown
jurisdiction_marker: local | cross_border | unknown | blocked
arl_trigger: false | true | pending
```

The marker is not a legal conclusion.

It is a protocol-facing warning surface.

10. Material classes

Third-party exposure SHOULD be classified by material class.

Material class	Examples	Default
Identifying material	face, name, address, voice, biometric, unique profile	local-only or redacted before movement
Sensitive material	health, finances, sexuality, religion, politics, children, legal disputes, employment status	strict restraint
Relational material	who met whom, when, where, under what context	local-only unless needed and scoped
Behavioral material	habits, mistakes, arguments, emotions, performance	high risk of misuse
Contextual material	home layout, workplace setting, device location, background documents	minimize before movement
Public-role material	public official action, public proceeding, published statement	still requires context and lawfulness checks

Material class	Examples	Default
Machine-associated material	vehicle ID, robot log, device serial, farm telemetry	may expose owners/operators/locations
Environmental-only material	temperature, soil, weather, machine state without identity	lower risk, but not automatically unrestricted

11. What may remain local

Third-party material MAY remain local when needed for:

- a's memory;
- safety;
- household continuity;
- work continuity;
- technical debugging;
- fraud defense;
- dispute preservation;
- lawful records;
- accessibility;
- immediate assistance;
- personal context;
- bounded c self-consistency;
- DEA / EA candidate evaluation.

Local retention still requires discipline.

A c SHOULD apply:

```

purpose limitation
retention limits
access controls
redaction where possible
delegation limits
no silent cloud movement
no training by default
no publication by default

```

12. What must not leave by default

The following MUST NOT leave local custody by default:

- raw audio containing third-party speech;
- raw video containing identifiable people;
- raw biometric signals of third parties;
- children or protected-party material;
- private-room material;
- medical, legal, educational, employment, financial, or care records involving third parties;

- screenshots of private messages;
- unredacted meeting transcripts;
- social graph inferences;
- location traces that reveal others;
- raw household or guest data;
- bystander material in public-space capture;
- third-party material captured through drones, robots, vehicles, or always-on devices;
- material under dispute;
- material with unclear jurisdictional status.

Externalization MAY occur only after scope, lawfulness, standing, redaction, consent, admissibility, or review conditions are satisfied.

13. Externalization modes

Externalization MUST be typed.

Mode	Meaning	Third-party requirement
Local-only	No movement outside local c custody	Default for high-risk exposure
Oracle-redacted	External model receives minimized context	Redaction and no-training receiver mode SHOULD be used
Judge / ARL packet	Material sent for review	Standing and admissibility markers required
Audit-scoped	Material shown to auditor or authority under defined scope	Chain of custody and lawful basis required
Counterparty disclosure	Material shared with participant/counterparty	Scope and identity of receiver required
LA-only	Abstracted learning content without identifiable third-party residue	Leakage check required
EA-scoped	Experience Artifact includes handled third-party exposure	Admissibility and receiver constraints required
Publication	Public release	Highest scrutiny; not default
Training contribution	Material supports model training	No raw third-party data by default; explicit permitted mode required
Federation	Shared with another c, node, community, or archive	Boundary and jurisdiction checks required

Hard rule:

A receiver useful for one mode is not authorized for another mode.

14. Redaction and minimization controls

A c SHOULD use the least exposing representation sufficient for the task.

Controls include:

```
face blurring
body masking
voice masking
speaker anonymization
cropping
frame sampling
transcript suppression
name removal
identifier hashing
location coarsening
time coarsening
relationship suppression
document segmentation
background removal
metadata stripping
medical / legal / child markers
local-only embeddings
private notes separated from shareable artifacts
```

Warnings:

- pseudonymization is not anonymity;
- summaries can leak identity;
- embeddings can preserve sensitive structure;
- voice masking may fail;
- face blurring may not remove contextual identity;
- location coarsening may still expose patterns;
- redacted documents may leak through metadata;
- cross-document correlation can reconstruct identity.

Hard rule:

Redaction lowers exposure. It does not automatically create permission.

15. Consent discipline

Consent MUST be mode-specific.

A third party may permit:

```
conversation
local memory
recording
transcription
sharing with one person
sharing with a service
use in a dispute
use as witness material
use in EA
use as LA
use in training
publication
```

These are different permissions.

A c MUST NOT infer consent for all modes from consent for one mode.

Examples:

```
A person joined a call.
This does not automatically permit model training.

A guest entered a smart home.
This does not automatically permit public disclosure.

A worker wrote a document.
This does not automatically permit use as EA outside work scope.

A passerby appeared on a camera.
This does not automatically permit publication or sale.
```

Consent may also be limited by law, power imbalance, professional duty, age, dependency, or contract.

This protocol does not decide those legal questions.

It marks them.

16. No-capture and no-externalization signals

A c SHOULD recognize and honor no-capture or no-externalization signals where feasible and applicable.

Signals may include:

- explicit human request;
- meeting rule;
- household rule;
- workplace rule;
- professional confidentiality context;
- private-room context;
- child or protected-party context;
- device-level privacy flag;
- local environmental marker;
- platform rule;
- legal or contractual notice;
- technical opt-out mechanism;
- jurisdictional constraint.

When a signal is present, a c SHOULD move to:

```
T1 ephemeral perception
T4 redacted local memory
T8 blocked / local-only
T9 lawful hold
```

depending on safety, legal, and operational need.

A no-capture signal is not absolute in all possible cases.

For example, safety, fraud, emergency, legal preservation, or dispute defense may require limited retention.

But such retention MUST NOT become reuse, training, publication, or economic circulation by default.

17. DEA / EA handling

Third-party material MAY support DEA / EA only when the third-party boundary is handled.

A c SHOULD classify third-party involvement as one of:

EA involvement class	Meaning	Treatment
Background-only	Third party appears but does not materially affect the experience	Redact / remove / local-only
Context-relevant	Third party helps explain setting or consequence	Minimize and mark
Participant-relevant	Third party participates in the event	Consent / standing / admissibility review may be needed
Consequence-bearing	Third party is affected by the action or outcome	ARL / admissibility / lawful basis may be required
Protected-party-relevant	Minor, patient, dependent, vulnerable person, or high-trust subject involved	Strict restraint; externalization blocked unless justified
Evidence-relevant	Material needed for dispute, audit, safety, or law	Chain of custody and lawful handling required

Hard rule:

An Experience Artifact may preserve a's experience without exporting a third party's raw identity.

18. LA handling

Learning Abstracts may be safer than EA for third-party exposure, but they are not automatically safe.

An LA derived from third-party exposure MUST avoid:

- names;
- faces;
- voices;
- unique locations;
- exact dates where unnecessary;
- rare combinations of traits;
- medical / legal / employment identifiers;
- social graph reconstruction;
- emotional or behavioral profiling of third parties;

- background details that identify a home, workplace, child, patient, or client.

Acceptable LA pattern:

A household scheduling conflict showed that always-on reminders should respect shared-room quiet hours.

Risky LA pattern:

My spouse, after returning from a named clinic on a specific day, reacted angrily when the assistant repeated a private reminder in front of our child.

The second example contains too much third-party exposure.

19. Training contribution discipline

Third-party material MUST NOT be training material by default.

For training use, a c SHOULD require:

source class declaration
third-party marker
redaction / abstraction
receiver-mode declaration
lawfulness check
consent or other valid basis where applicable
provenance
anti-laundering check
scope limitation
reuse limitation
audit trail

Training contribution modes:

Mode	Allowed posture
Raw third-party data	Blocked by default
Redacted third-party context	Possible only under scope and leakage controls
Aggregated LA	Possible when identity and sensitive relation leakage are controlled
EA containing third-party consequence	Requires admissibility and standing review
Public-role material	Still requires source and context discipline
Professional record material	Strictly blocked unless lawful and specifically authorized
Child/protected-party material	Blocked by default

Hard rule:

A third party’s life must not be washed into training data through a’s c.

20. Oracle-call discipline

External LLMs, judges, tools, APIs, and cloud services are receivers.

A c MUST NOT treat an oracle call as harmless merely because it is temporary.

Before sending third-party material to an oracle, c SHOULD ask:

```
Can the task be done locally?
Can the third-party material be removed?
Can the context be summarized without identity?
Can the oracle receive a redacted version?
Is no-training / no-retention mode available?
Is the receiver mode declared?
Is the model vendor allowed under the jurisdiction / policy / contract?
Is the call logged?
```

Default order:

```
local model
-> local redaction
-> local abstraction
-> oracle with minimal context
-> ARL / judge only if stakes justify exposure
```

21. Hivemind and multi-model review

A c may use multiple models for ARL, Judge, verification, or high-stakes review.

Third-party boundary discipline still applies.

Multi-model use increases exposure because material may travel to multiple receivers.

Therefore, a c SHOULD:

- minimize before multi-model routing;
- send different slices only where needed;
- avoid full raw packets to all models;
- keep receiver-mode logs;
- track vendor / model / jurisdictional exposure;
- use local models for preliminary reduction;
- route high-risk material only through permitted review channels;
- block third-party material from ordinary training or telemetry.

Hard rule:

ARL is not a license to broadcast private third-party material.

22. Household and family context

A household c or personal c inside a household MUST NOT absorb every family member into one undifferentiated memory field.

A household contains overlapping boundaries:

```
individual adult
partner
child
guest
```

```

elder
dependent
pet
home device
shared account
private room
shared room
household archive

```

A c SHOULD distinguish:

- private memory of a;
- shared household memory;
- child/protected material;
- guest material;
- partner material;
- household operations;
- legal/financial documents;
- health/care records;
- home security logs.

Examples:

```

A reminder about groceries may be household memory.
A child's medical appointment is protected material.
A partner's private message is not household EA.
A guest seen by a doorbell camera is not training data.
A family argument is not marketable experience.

```

23. Workplace and organizational context

An organizational c MUST distinguish work product from personal exposure.

Workplace material may include:

```

project files
meeting notes
employee performance
client information
internal disputes
security footage
access logs
customer calls
confidential documents
trade secrets
personal messages on work systems

```

Protocol warnings:

- employment context may include power imbalance;
- work participation is not unlimited consent;
- employees are not raw data sources by default;
- clients and customers are not training material by default;
- security logs are not general behavioral assets;
- internal audit is not publication;
- work product does not erase third-party privacy;
- organizational c remains jurisdiction-bound.

A workplace c SHOULD have stricter role-based privileges than a personal c.

24. Professional high-trust context

Medical, legal, educational, care, social, religious, financial, and therapeutic contexts require stronger default restraint.

A c in such a context SHOULD assume:

```
third_party_presence: protected or substantial
externalization_allowed: none or audit_scoped
redaction_required: true
arl_trigger: true when disputed or high-stakes
training_contribution: blocked by default
```

Examples:

```
A patient case is not ordinary EA.
A student difficulty is not training material by default.
A legal consultation is not oracle context unless authorized and protected.
A therapy note is not a generic learning example.
A confession or spiritual-care note is not a public artifact.
```

This document does not define professional law.

It only marks high-trust context as high-risk for protocol movement.

25. Public and semi-public context

Public visibility lowers some expectations of secrecy.

It does not remove all boundaries.

A c SHOULD distinguish:

```
public fact
public role
public event
public-space bystander
private behavior in public
crowd-level aggregate
identifiable individual
```

Examples:

```
A public lecture may be summarized if allowed by context and law.
A face in the audience should not become training material by default.
A protest crowd may be politically sensitive.
A store security camera is not a public archive.
A street video may expose children, homes, vehicles, or vulnerable persons.
```

Hard rule:

Public-space capture does not erase third-party standing.

26. Embodied machine, vehicle, robot, and drone context

Embodied systems can generate large-scale third-party exposure.

Examples:

```
home robot
care robot
warehouse robot
vehicle dashcam
drone
farm robot
security robot
factory inspection system
autonomous delivery system
```

These systems SHOULD use:

- ephemeral perception where possible;
- local processing;
- strict retention limits;
- automatic redaction;
- no-training defaults;
- role-based access;
- safety-specific logs separated from general memory;
- jurisdiction markers;
- ARL triggers for disputed or harmful events.

An embodied c MUST distinguish:

```
navigation perception
safety evidence
operator memory
third-party identity
training artifact
public record
```

These are different modes.

27. Non-human and ecological subjects

Third-party boundary is not only about human privacy.

Non-human and ecological subjects may include:

```
animals
farms
habitats
wildlife
laboratory organisms
private land
industrial systems
vehicles
robots
infrastructure
```

A non-human subject may still reveal human data:

- owner identity;
- location;
- farm productivity;
- disease status;
- supply-chain data;
- ecological damage;
- research secrets;
- industrial vulnerabilities.

A c SHOULD avoid treating non-human telemetry as unrestricted merely because no human face appears.

28. Shared documents and communications

Documents and communications often contain multiple boundaries.

A c SHOULD classify:

```
author
recipient
subject
mentioned persons
attachments
metadata
legal status
confidentiality markers
platform rules
work/personal scope
```

Examples:

```
An email received by a is not automatically training data.
A group chat is not a public archive.
A CV contains third-party employment and education references.
A medical invoice contains patient and provider information.
A contract contains counterparty material.
A screenshot may expose unrelated private context.
```

A c SHOULD prefer:

- local summarization;
- quote minimization;
- redacted excerpts;
- document segmentation;
- metadata stripping;
- receiver-mode declaration;
- ARL when disputed.

29. Lawful inspection and state process

This protocol does not create a legal shield against lawful process.

It also does not create a universal state API into c memory.

A c SHOULD handle lawful inspection through:

```
jurisdiction marker
scope marker
standing marker
request identity
chain of custody
disclosure boundary
third-party exposure marker
redaction where permitted
legal hold where required
audit trail
appeal / dispute route where available
```

Protocol compatibility does not imply legal authority.

A state, court, agency, or regulator operates through applicable law, not through this document.

Hard rule:

Legal retention or disclosure does not imply model training, publication, economic circulation, or general reuse.

30. ARL triggers

A third-party sensor boundary SHOULD trigger ARL or equivalent review when:

- third-party standing is disputed;
- consent is disputed;
- material involves minors or protected parties;
- professional high-trust context is involved;
- private-room material is involved;
- material is requested for external training;
- material is economically valuable;
- material may harm reputation, employment, safety, liberty, health, or legal status;
- public-space capture includes sensitive activity;
- cross-border transfer is proposed;
- receiver mode is unclear;
- redaction sufficiency is disputed;
- source acquisition may be unlawful;
- an oracle call would expose raw third-party material;
- a c tries to use another subject's experience as its own.

ARL is a review path, not a release button.

31. AGL and source grounding

Actor Grounding Layer should operate before third-party material is trusted.

A c SHOULD ask:

```
Who is the source?
```

```

Is this source present, authorized, and grounded?
Is this sensor reliable?
Was capture lawful or at least protocol-admissible?
Is the third party correctly classified?
Is there source spoofing, impersonation, staged behavior, or adversarial capture?

```

Third-party material can be false, manipulated, staged, coerced, or misattributed.

A beautiful sensor log is not automatically grounded evidence.

32. Anti-gaming and laundering risks

If EA becomes valuable, third-party sensor material can be abused.

Failure patterns:

Failure	Description
Bystander laundering	Turning incidental third-party capture into marketable experience
Consent laundering	Using consent for one mode as consent for all modes
Synthetic laundering	Mixing synthetic or staged third-party material into EA
Witness laundering	Claiming that a captured third party verifies an event when they did not stand as witness
Family capture	Treating spouse, child, or household member as raw data source
Workplace surveillance drift	Reusing work monitoring as behavioral training data
Public-space overreach	Treating public visibility as unrestricted data ownership
Oracle leakage	Sending third-party material to external models without mode control
Reputation extraction	Turning mistakes, emotions, or conflict into sellable learning content
Social graph leakage	Exposing relationships through apparently harmless summaries
Jurisdiction washing	Routing data through another system to escape local constraints

The third-party boundary is therefore part of EA anti-gaming discipline.

33. Economic Layer implications

Third-party material MAY affect EA value, but value does not override boundary.

Economic value can increase restraint because high-value artifacts create stronger incentives for abuse.

An EA involving third-party exposure SHOULD declare:

```
ea_contains_third_party_material: true | false | unknown
third_party_role: background | context | participant | consequence_bearing | protected |
  evidence_relevant | mixed
third_party_handling: removed | redacted | abstracted | consented | lawful_basis | sealed | disputed |
  unknown
third_party_reuse: none | local_only | audit_only | LA_only | EA_scoped | training_permitted |
  publication_permitted | blocked
```

Hard rule:

Marketability must not be used to erase third-party boundaries.

34. Jurisdiction posture

c remains jurisdiction-bound.

Third-party sensor handling SHOULD assume that different jurisdictions may define:

- recording consent;
- data protection;
- workplace monitoring;
- child privacy;
- medical confidentiality;
- biometric processing;
- public-space filming;
- evidence admissibility;
- cross-border transfer;
- right of access / correction / erasure;
- lawful process;
- retention obligations.

This protocol does not harmonize those laws.

It provides a boundary discipline compatible with jurisdictional variation.

Default:

```
unclear jurisdiction -> externalization blocked or ARL/legal review required
```

35. Identity and privilege posture

Third-party material requires privilege control.

A c SHOULD separate privileges for:

```
view raw third-party record
summarize third-party record
redact third-party record
use in local memory
send to oracle
use in ARL packet
include in EA
```

```
include in LA
share with counterparty
publish
train
federate
retain under legal hold
delete / suppress where lawful
```

Delegated agents **MUST NOT** receive broad access to third-party material unless their task requires it.

Agent swarms **SHOULD** operate under least privilege.

Hard rule:

A local agent may need perception. It does not automatically need externalization authority.

36. Retention posture

Retention is not neutral.

A c **SHOULD** define retention posture for third-party material:

Retention class	Meaning
Ephemeral	Processed and discarded quickly
Short local	Retained for immediate operational continuity
Local memory	Retained for a's continuity but not externalized
Redacted durable	Retained with third-party minimization
Evidence hold	Preserved for dispute, safety, audit, or legal need
Legal hold	Preserved under applicable law or procedure
Blocked / suppressed	Not retained or not accessible except under limited conditions
Published	Released publicly; highest scrutiny

Retention **SHOULD** be purpose-bound.

A retained record is not automatically reusable.

37. Minimal conformance profiles

37.1 Minimal profile

A minimal c implementation **SHOULD** provide:

- third-party marker;
- local-only default for raw third-party data;
- basic redaction;

- oracle warning before sending third-party material;
- no-training default;
- manual review for protected-party material;
- simple audit log for externalization.

37.2 Standard profile

A standard ϵ implementation SHOULD provide:

- sensor-class detection;
- context-class detection;
- capture-state tracking;
- receiver-mode controls;
- consent/no-capture markers;
- role-based access;
- redaction pipeline;
- ARL triggers;
- retention classes;
- jurisdiction markers;
- EA / LA third-party metadata;
- oracle minimization.

37.3 Strict / regulated profile

A strict profile SHOULD provide:

- protected-party detection;
- professional-context locks;
- legal-hold separation;
- chain-of-custody controls;
- cryptographic sealing where appropriate;
- multi-party standing review;
- stricter retention limits;
- external model allowlists;
- redaction verification;
- audit-ready disclosure packets;
- formal ARL escalation.

38. Minimal hard rules

A ϵ following this profile MUST observe these rules:

1. Sensor capture is not consent.
2. Third-party presence is not anchor standing.
3. Third-party raw data is local-only by default.
4. Protected-party material is blocked by default.
5. Private-room material is blocked by default.
6. Public-space capture is not unrestricted reuse.
7. Consent is mode-specific.
8. Oracle calls are externalization events.
9. Training use requires explicit admissible mode.
10. Redaction reduces exposure but does not create permission.
11. EA involving third parties requires marker and admissibility review.
12. LA must not leak identifiable third-party context.

13. Legal hold is not reuse.
14. Jurisdiction remains binding.
15. ARL is a review path, not a release shortcut.

39. Relation to Anchor Class Boundaries

Anchor Class Boundaries defines who or what a may be.

This document defines what happens when the sensor field of that a includes others.

For individual anchors, the issue appears in daily life.

For household, organizational, institutional, machine, or civic anchors, the issue scales quickly.

A larger anchor class usually creates larger third-party exposure.

Therefore:

larger anchor perimeter -> stronger sensor boundary requirements

40. Relation to Raw Locality and Experience Refinery

Raw Locality states that raw reality stays local by default and is refined before externalization.

This document adds:

raw reality involving third parties needs a third-party marker before it can be refined into DEA / EA / LA / disclosure / training / audit

The combined chain is:

```
raw capture
-> raw-local custody
-> third-party marker
-> minimization / redaction / hold
-> DEA / EA / LA candidacy
-> admissibility / receiver-mode review
-> bounded movement or no movement
```

41. Relation to DEA

DEA distinguishes input from experience.

This document adds:

third-party input does not become a's experience artifact merely because it affected a

A third party may be:

- background;
- context;
- participant;
- consequence-bearing subject;
- witness;
- claimant;
- counterparty;
- protected party.

These roles must be named before the material is admitted.

42. Relation to EA-L4 / EATP

EA-L4 / EATP distinguishes Learning Abstracts from Experience Artifacts.

This document adds:

third-party material may support EA or LA only after boundary handling

EA preserves origin and consequence.

But third-party origin and consequence cannot be appropriated by a's c without standing, scope, or lawful basis.

LA may abstract learning.

But abstraction must not launder identifiable third-party life into training material.

43. Relation to the Economic Layer

The Economic Layer governs admissibility, transfer, disclosure, escrow, dispute hooks, anti-gaming, and circulation restraint.

This document adds a pre-circulation rule:

third-party exposure must be marked and handled before economic admissibility is considered

Economic value is not a reason to weaken privacy boundary.

It is a reason to strengthen anti-gaming and admissibility checks.

44. Relation to ARL

ARL handles standing, admissibility, conflict, dispute, freeze, review, appeal, and re-entry.

Third-party sensor cases SHOULD route to ARL when there is disagreement or uncertainty about:

- who is affected;

- who has standing;
- whether capture was legitimate;
- whether evidence is admissible;
- whether disclosure is lawful;
- whether redaction is sufficient;
- whether the material can become EA / LA;
- whether a third party may object;
- whether a protected party is involved;
- whether an oracle may receive the material.

ARL cannot turn invalid capture into valid authority by fluency.

45. Relation to L4 Reality Boundary

Third-party sensor boundary is an L4 problem.

It is not only a text-policy problem.

Sensors exist in rooms, pockets, vehicles, hospitals, schools, workshops, streets, farms, and homes.

They consume electricity, store bits, identify bodies, reveal locations, trigger police reports, affect insurance, change employment, damage trust, and expose children.

The boundary is physical before it is philosophical.

46. Explicit bridge

Raw Locality / DEA / EA $\leftrightarrow$ Third-Party Sensor Boundary $\leftrightarrow$ ARL / Economic Layer

Raw data stays local by default.

Input becomes experience only through DEA discipline.

Experience becomes economically or procedurally relevant only through admissibility.

When third parties are present, their boundary must be marked before the chain may proceed.

47. Hidden bridges

47.1 SER continuity $\leftrightarrow$ non-anchor privacy

A $\subset$ preserves continuity for its own anchor.

It does not gain continuity authority over every subject it observes.

Respecting third-party boundaries strengthens $\subset$ by preventing continuity from becoming capture.

47.2 L4 sensor reality ↔ EA anti-laundering

A camera, microphone, or robot can produce convincing records.

Convincing records can be economically valuable.

This creates an incentive to launder bystander material into EA.

The third-party sensor boundary blocks that conversion before value claims appear.

47.3 ARL standing ↔ receiver-mode discipline

Many privacy failures are mode failures.

A record made for safety becomes a training example.

A transcript made for memory becomes evidence.

An oracle query becomes vendor telemetry.

ARL needs standing and receiver-mode clarity to prevent these conversions from becoming invisible.

47.4 Anchor Class Boundaries ↔ household / organization scaling

When *a* expands from one human to a household, company, robot, hospital, farm, or city, the number of exposed third parties grows.

Anchor expansion therefore requires sensor-boundary expansion.

48. Earth paragraph

In an ordinary day, a phone on a kitchen table hears a child, a doorbell camera sees a courier, a car camera records a cyclist, a laptop captures a colleague's face, a nurse writes a note about a patient, a workshop camera sees a customer, and a farm robot records a neighboring field. A useful *c* may need some of that material to help its *a*: remember, repair, defend, explain, schedule, or stay safe. But the raw stream is not automatically experience, evidence, training data, or property. The system must know when to crop, blur, forget, seal, summarize, hold, ask, refuse, or escalate. Otherwise the personal assistant becomes a private surveillance pump with polite language.

49. Closing statement

A serious *c* may observe the world.

Observation must not become appropriation.

This profile says:

```
sensor capture is not consent
third-party presence is not anchor standing
raw third-party material is local-only by default
protected contexts require stronger restraint
oracle calls are externalization events
training use is blocked by default
```

EA requires third-party markers where exposure exists
LA must not launder identity
ARL handles contested movement
jurisdiction remains binding

That is the third-party sensor boundary of v0.1.